Structure-informed CRISPR co-dependency maps the ARID1A-associated BAF module across cancer cell lines

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Abstract

ARID1A is recurrently inactivated in non-small cell lung cancer, but the functional organization of the ARID1A-associated BAF complex is incompletely resolved. We applied five methodologically distinct analyses to genome-scale CRISPR dependency profiles. All converged on positive co-dependency between ARID1A and core-shared or cBAF-associated subunits, whereas PBAF- and ncBAF-specific subunits were null or negative. The pattern was robust across 19 tissues, two processing resources, and explicit pan-essentiality and lineage controls. In the assembled human BAF structure, ARID1A-contacting subunits had larger co-dependency correlations than non-contacting or mutually exclusive subunits (lung median difference 0.436, 95% bootstrap interval 0.293–0.584; Cliff's $\delta = 0.86$, exact $P = 0.0037$). Contact count did not show a positive dose–response among the six contacts resolved in the same structure, defining structural contact as a binary prior rather than a quantitative predictor. Restricting the core module to load-bearing members improved its anchor-excluded correlation. In a symmetric nested analysis of 266 curated complexes, with both anchor and dropped member selected only in the discovery half, held-out drop-identity concordance exceeded that of size-matched null modules by a median 0.25 (95% bootstrap interval 0.175–0.308; paired $P = 1.0 \times 10^{-19}$). Mutation-, protein- and mRNA-defined ARID1A-low tests did not survive multiple-testing correction, despite detection of a prespecified SMARCA4-low–SMARCA2 positive control. The results therefore map co-essential organization and do not establish an ARID1A-loss synthetic-lethal vulnerability. This distinction provides a rigorous framework for using CRISPR co-dependency to refine protein-complex modules.

Keywords: ARID1A; SWI/SNF (BAF) complex; non-small cell lung cancer; co-dependency; co-essentiality; protein complex; chromatin remodeller

1 Introduction

The SWI/SNF (BAF) chromatin-remodelling complex carries an inactivating mutation in at least one subunit in roughly 20% of human cancers [1], and is now an actively pursued therapeutic

target class [2, 3]. It assembles modularly into three mutually exclusive forms — canonical BAF (cBAF), polybromo-associated BAF (PBAF) and non-canonical BAF (ncBAF) — sharing a set of core subunits and each carrying defining subunits [4–6]. These defining subunits are biologically mutually exclusive: a single complex never contains both ARID1A and PBRM1. This exclusivity is the interpretive basis for the structural analysis below.

ARID1A defines cBAF and is recurrently inactivated in NSCLC [7], where SWI/SNF alterations are associated with poorer immunotherapy outcomes [8]. The lesion itself is not directly druggable, and genetic-interaction studies have therefore sought dependencies associated with the ARID1A-mutant state [9]. A separate and more general question is how ARID1A dependency covaries with other subunits of the assembled BAF complex across heterogeneous cancer cell lines. This continuous co-dependency describes co-essential organization; it is not equivalent to a genotype-conditional synthetic-lethal effect.

Two paradigms motivate different genetic tests. The paralogue synthetic-lethality paradigm holds that ARID1A-mutant tumours depend on ARID1B [10], a relationship since generalised into systematic paralogue-dependency maps [11, 12]. The residual-complex compensation paradigm holds instead that cBAF loss drives compensatory upregulation of ncBAF/PBAF and a switch in dependency to those residual complexes [13]. The two point to different target sets, but loss-conditioned dependency and continuous co-dependency must be analysed and interpreted separately.

Any study of complex-specific dependency also confronts three standing objections: **circularity** (pre-specifying a subunit list makes "discovering" a complex dependency vacuous), the **pan-essentiality artefact** (essential genes covary for reasons unrelated to shared membership), and **lineage confounding** (cell lines cluster by tissue, and module dependencies vary by tissue) [14–16].

A fourth problem receives almost no attention. Module-based analysis requires a *definition* of the module, inherited from biochemistry and used as a fixed input [17]. Nobody asks whether it is the right definition for a dependency analysis: a subunit may be a bona fide member by co-purification yet contribute nothing to — or actively dilute — the dependency signal, because it does not contact the anchor or because paralogues cover its loss.

We address all four. Without pre-specifying any gene list, five methodologically distinct genome-scale analyses are required to converge; pan-essentiality and lineage controls are built in; and the curated module definition is tested rather than assumed. These analyses share a dependency matrix and are therefore convergent views of one functional-genomic measurement, not five independent biological experiments. The experimental structure supplies an orthogonal test and nominates candidate members for refinement; we show that its second role is heuristic rather than decisive. Figure 1 sets out the complete evidence architecture, including the orthogonal structural test, replication, cross-layer checks and the bounded therapeutic analyses that are kept outside the primary evidentiary chain.

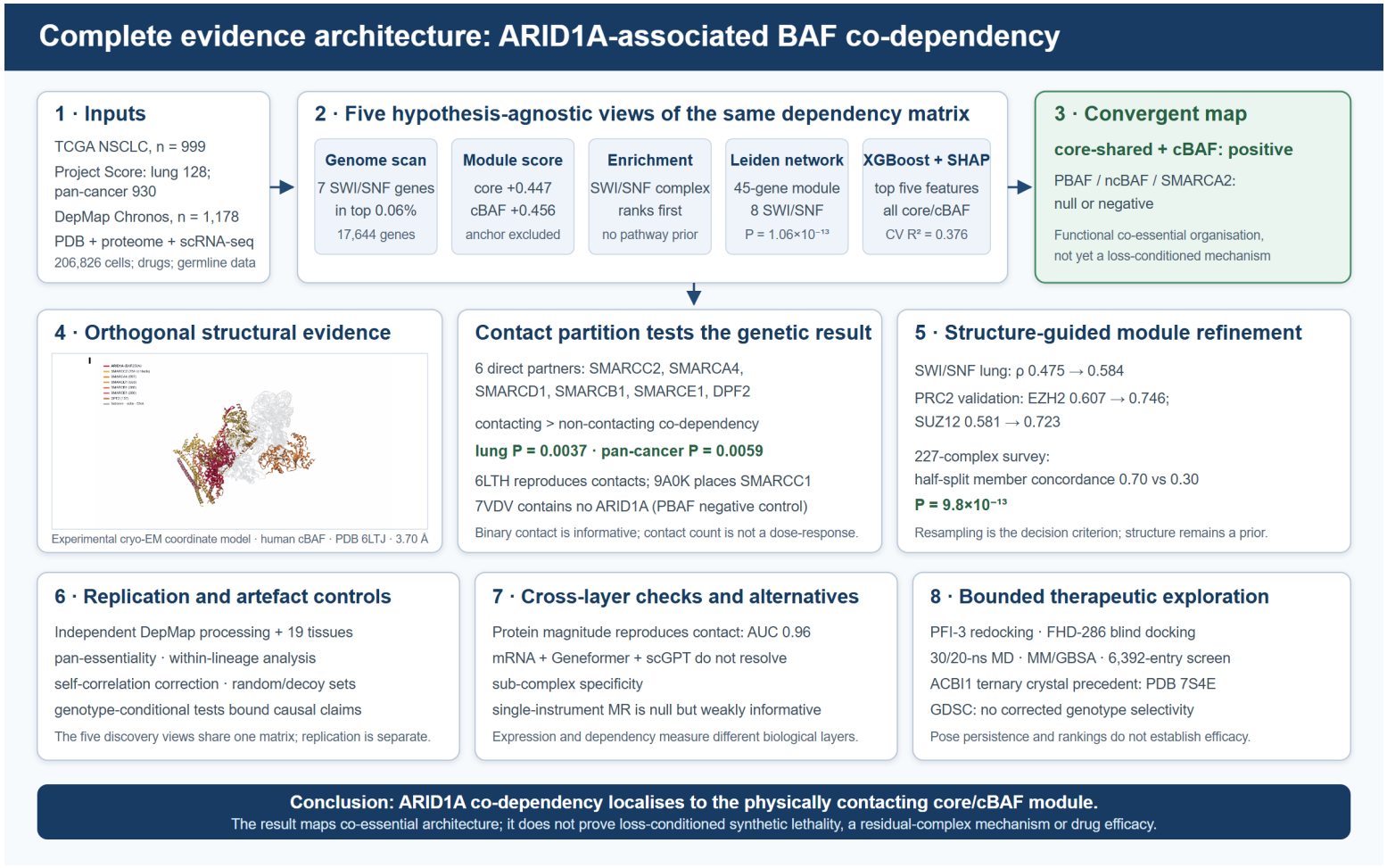


Figure 1: Complete evidence architecture. Five genome-scale views converge on the core/cBAF module; experimental structure, module refinement, replication and cross-layer checks then test and bound the result. The structural thumbnail renders the experimentally determined cryo-EM coordinate model (PDB 6LTJ), not a raw micrograph or density map. Therapeutic analyses remain outside the primary evidence chain.

2 Results

2.1 ARID1A is recurrently mutated and frequently truncated in NSCLC

In the combined TCGA-LUAD and TCGA-LUSC cohort ($n = 999$) [18], *ARID1A* was mutated in **6.51% of cases (65/999)**, at closely similar frequencies in the two histologies (LUAD 6.23%, 32/514; LUSC 6.80%, 33/485; Figure 2a), establishing ARID1A as a recurrently altered gene in this disease context. The rate replicates outside TCGA: two independent lung adenocarcinoma cohorts queried through the cBioPortal API [19, 20] give 5.38% (10/186, MSK-IMPACT [21]) and 6.67% (177/2,653 [22]) — both within a point of the TCGA estimate despite different platforms, populations and ascertainment.

The mutational spectrum is that of a tumour suppressor (Figure 2b): of 73 variants, **54.8% (40/73) were truncating**, distributed without hotspots, against ∼68% for *STK11* in the same cohort and almost entirely hotspot missense for *KRAS*. The spectrum is consistent with frequent loss of function, although mutation status alone does not establish complete loss of ARID1A protein in an individual model.

Stratifying TCGA-LUAD by *ARID1A* expression tertile (low $n = 180$ versus high $n = 180$), DESeq2 [23] detected **501 significant genes** (adjusted $P < 0.05$, $|\log_2 \mathrm{FC}| > 1$; 421 up, 80 down; Figure 2c). This associative contrast establishes a transcriptional context for low ARID1A expression but is not interpreted as a causal perturbation.

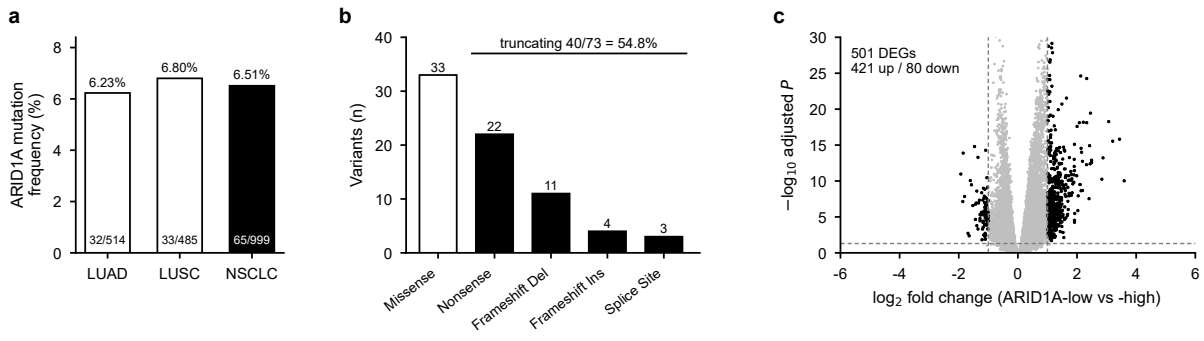


Figure 2: ARID1A is recurrently mutated and frequently truncated in NSCLC. (**a**) Mutation frequency in TCGA-LUAD, TCGA-LUSC and the combined NSCLC cohort; counts give mutated cases over cohort size. (**b**) Variant classification spectrum; filled bars are truncating classes. (**c**) Volcano plot of ARID1A-low versus ARID1A-high tertile differential expression; filled points pass adjusted $P < 0.05$ and $|\log_2 \mathrm{FC}| > 1$.

2.2 Five methodologically distinct analyses converge on the ARID1A-associated BAF module

Five analyses were run separately on the same dependency matrix; four of the five use no complex composition at all (Figure 3). Supplementary Figure S1 gives the sub-complex compositions and the structure of the convergence argument.

Genome-scale scan. Correlating the ARID1A dependency profile from the Project Score screen [24, 25] against ∼17,644 genes in lung cancer ($n = 128$), the top-ranked genes were SMARCE1 (#1, $\rho = 0.551$, $P = 1.6 \times 10^{-11}$), SMARCB1 (#2, 0.474), SMARCC1 (#3, 0.380), DPF2 (#4, 0.363), SMARCA4 (#5, 0.357), ARID1B (#7, 0.344) and SS18 (#10, 0.313) — seven SWI/SNF subunits inside the top 0.06% of the genome (hypergeometric enrichment in the top 1%: $P = 3.2 \times 10^{-12}$ lung, 5.5×10^{-14} pan-cancer), while PBAF- and ncBAF-specific subunits ranked in the bottom 90–97%. The pan-cancer ranking has the same shape (Supplementary Figure S2): eight of the top nine genes are SWI/SNF subunits, with the paralogue ARID1B only 32nd. A network-guided druggability overlay places SMARCA4 first among co-dependencies for which a chemical probe or clinical-stage inhibitor exists. This overlay prioritises follow-up targets; it is not a compound–target–pathway network-pharmacology analysis and does not establish genotype-selective efficacy.

Module scoring. Core-shared and cBAF-specific modules were significantly positive at both levels; PBAF-specific and alternative-ATPase modules null; the ncBAF-specific module significantly negative pan-cancer (Table 1).

Table 1: Complex-module co-dependency with ARID1A. The anchor is excluded from every module (Section 2.7).

Module	**Subunits**	**Lung** ρ	P	**Pan-cancer** ρ	P
Core-shared	8	+0.447	1.2×10^{-7}	+0.461	3.1×10^{-50}
cBAF-specific	9	+0.456	6.2×10^{-8}	+0.307	1.0×10^{-21}
PBAF-specific	4	−0.123	0.166	−0.012	0.720
ncBAF-specific	3	−0.157	0.076	−0.110	8.1×10^{-4}
Alt. ATPase (SMARCA2)	1	−0.122	0.170	−0.002	0.958

Prior-free enrichment. The top-100 co-dependent genes, submitted to Enrichr [26] without context, returned SWI/SNF Complex (GO:0016514) as the top term in lung (9/29 genes, adjusted $P = 1.5 \times 10^{-12}$), followed by nBAF, CORUM EBAFb, npBAF and BAF — every leading term a SWI/SNF sub-complex.

Co-essentiality community detection. From 7,940 selectively essential genes we built a genome-wide k-nearest-neighbour co-dependency graph ($K = 15$, $\rho > 0.15$) and ran Leiden community detection [27] across a resolution sweep. At resolution 4.0 the ARID1A community contained **45 genes including 8 SWI/SNF subunits** (hypergeometric $P = 1.06 \times 10^{-13}$). The algorithm is given neither the anchor's identity nor the study question.

Gradient boosting with SHAP. Predicting ARID1A dependency from 3,000 genome-scale features (1,178 lines) gave cross-validated $R^2 = 0.376$. The top five SHAP features [28] were SMARCB1, SMARCE1, SMARCC1, ARID1B and DPF2, separated by a clear discontinuity from rank 6. PBAF/ncBAF subunits ranked far lower (BRD9 #30, ARID2 #108, BRD7 #224, PBRM1 #301). Because the original training driver was not retained, we performed a fully specified pre-submission reanalysis with unsupervised variance-based feature selection and out-of-fold prediction.

It recovered the same five leading features and a similar out-of-fold $R^2 = 0.390$, providing a reproducible sensitivity analysis rather than relying on the historical model output alone.

Mandatory controls. ARID1A is *selectively*, not pan-, essential — dependent (gene effect < -0.5) in 15.4% of lines and strongly dependent (< -1.0) in 1.3% — whereas pan-essential references are essential in 100% of lines, so its correlations cannot be a pan-essentiality artefact. Correspondingly ARID1A co-dependency is strong with core BAF and null with every pan-essential reference ($|\rho| \leq 0.070$) and with PBRM1 (+0.002). Lung estimates are computed within lineage; robustness to broader lineage composition is assessed by separate within-tissue analyses across 19 tissues below.

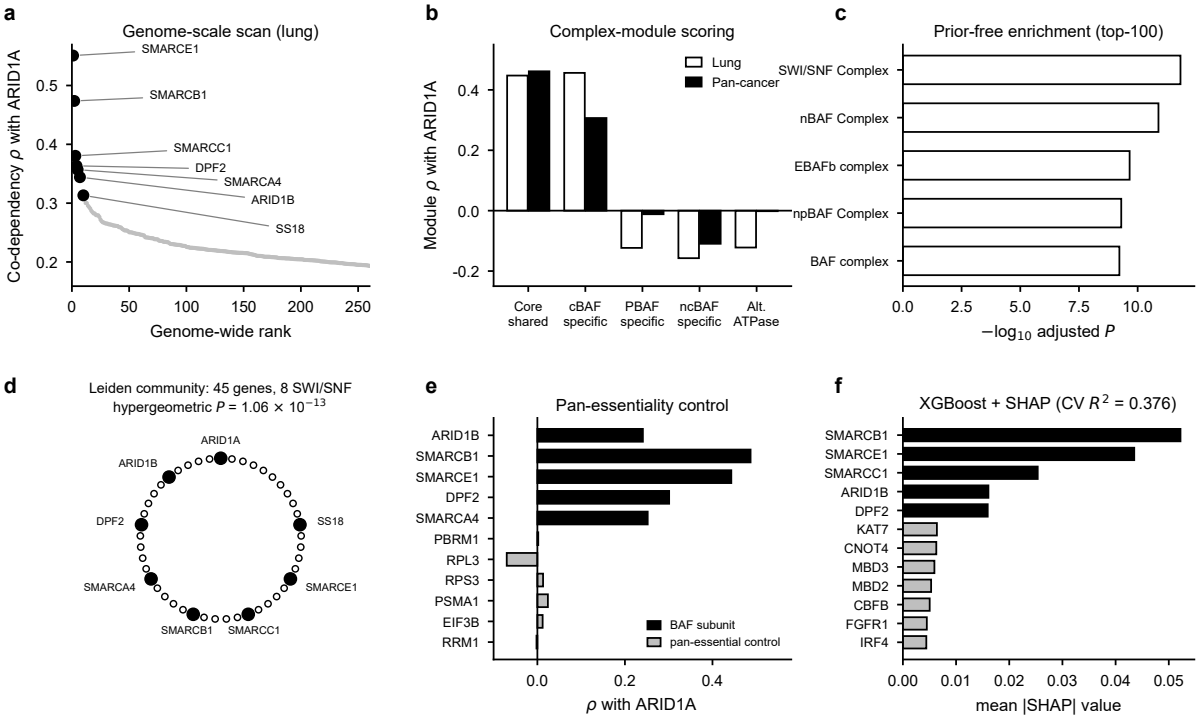


Figure 3: Five methodologically distinct analyses converge. (**a**) Genome-scale co-dependency scan in lung cancer; SWI/SNF subunits are highlighted. (**b**) Complex-module scores in lung and pan-cancer. (**c**) Prior-free enrichment of the top-100 co-dependent genes. (**d**) Leiden community containing ARID1A. (**e**) Pan-essentiality specificity controls. (**f**) XGBoost/SHAP feature attribution.

2.3 The experimental structure explains the genetic result and refines the module definition

The preceding section rests entirely on genetic correlation, so we introduced evidence of an entirely different kind: the cryo-EM structure of the assembled human BAF complex (PDB 6LTJ, 3.70 Å; base module 6LTH, 3.00 Å) [29], with the nucleosome-bound PBAF structure (PDB 7VDV) [30] as a negative control — it contains no ARID1A chain at all, confirming cBAF/PBAF mutual exclusivity structurally rather than by assumption.

Parsing chain coordinates with Biopython [31], ARID1A makes direct heavy-atom contact (minimum distance 2.20–2.47 Å) with SMARCC2 (754 atom pairs within 4.5 Å), SMARCA4 (557),

SMARCD1 (553), SMARCB1 (300), SMARCE1 (200) and DPF2 (137), and no contact with ACTL6A (13.02 Å) or ACTB (31.99 Å) (Figure 4I). SMARCC2 is present as a homodimer (chains N and O). Independent recomputation in the higher-resolution 6LTH gave identical counts to within one atom pair. SMARCC1, unresolved in 6LTJ, contacts ARID1A in the integrative model of the endogenous complex (PDB 9A0K; 186 pairs, 2.58 Å) [32]; the two agree on all six subunits 9A0K models to comparable depth, diverging only for SMARCE1 and DPF2, which it covers over 79 and 37 residues. We count SMARCC1 as contacting.

The wwPDB validation reports qualify how these coordinates should be used. The full 6LTJ deposit has a low global map–model Q-score (0.007; 0.6th percentile among same-resolution entries), whereas the independently deposited, higher-resolution 6LTH base module has Q=0.526 (67.5th percentile). We therefore use 6LTH as the primary replication of the interface rather than treating 6LTJ alone as definitive. All six contact calls persisted in both deposits at 4.0, 4.5 and 5.0 Å and after removing the bottom decile of residue Q-scores within every analysed chain (Supplementary Figure S8b). The 196 ARID1A interface residues were not selectively lower quality than non-interface ARID1A residues in either deposit (two-sided Mann–Whitney $P = 0.182$ for 6LTJ and $P = 0.169$ for 6LTH). Thus the binary interface assignment is deposit- and threshold-robust, although the low global quality of 6LTJ precludes stronger atom-by-atom mechanistic claims.

Physical contact separates stronger from weaker co-dependencies. Comparing the seven contacting genes against eight non-contacting or mutually exclusive ones gave a lung median difference in ρ of 0.436 (95% bootstrap interval 0.293–0.584; Cliff's $\delta = 0.857$; exact two-sided Mann–Whitney $P = 0.0037$) and a pan-cancer difference of 0.314 (0.165–0.451; $\delta = 0.821$; $P = 0.0059$). Restricted to the six subunits resolved in 6LTJ, the group comparisons gave $P = 0.0080$ and $P = 0.0127$. This binary structural partition is an orthogonal measurement to the genetic screen. It should not be interpreted as a dose–response: among the six contacting subunits resolved in the same structure, contact count was not positively associated with co-dependency (lung Spearman $\rho = -0.657$, $P = 0.156$; pan-cancer $\rho = -0.600$, $P = 0.208$). Physical contact is therefore used as a binary prior, not as a quantitative predictor of dependency strength.

The group comparison was also insensitive to the contact definition (Supplementary Figure S8c). Excluding integrative-model-only SMARCC1 gave lung and pan-cancer median differences of 0.432 (95% bootstrap interval 0.162–0.590; exact $P = 0.0080$) and 0.279 (0.090–0.471; $P = 0.0127$), respectively. All 15 leave-one-gene-out versions remained significant in each cohort ($P < 0.05$). Restriction to the eight subunits physically present in 6LTJ retained the direction but, with only two non-contacting comparators, was underpowered ($P = 0.286$ in each cohort).

Curated module definitions are not optimal. Restricting the core module to the subunits the structure shows to contact the anchor gives a better module: on DepMap (lung $n = 126$; pan-cancer $n = 1{,}178$) the correlation rises from 0.475 to 0.584 (lung) and 0.499 to 0.533 (pan-cancer) while dropping two of eight genes (Figure 5c). ACTB and ACTL6A are individually null (lung $\rho = 0.043$ and 0.122) and act as dilution terms: in lung, leave-one-out over those same eight (Figure 5b) gains +0.061 and +0.037 on their removal while removing a contacting subunit costs

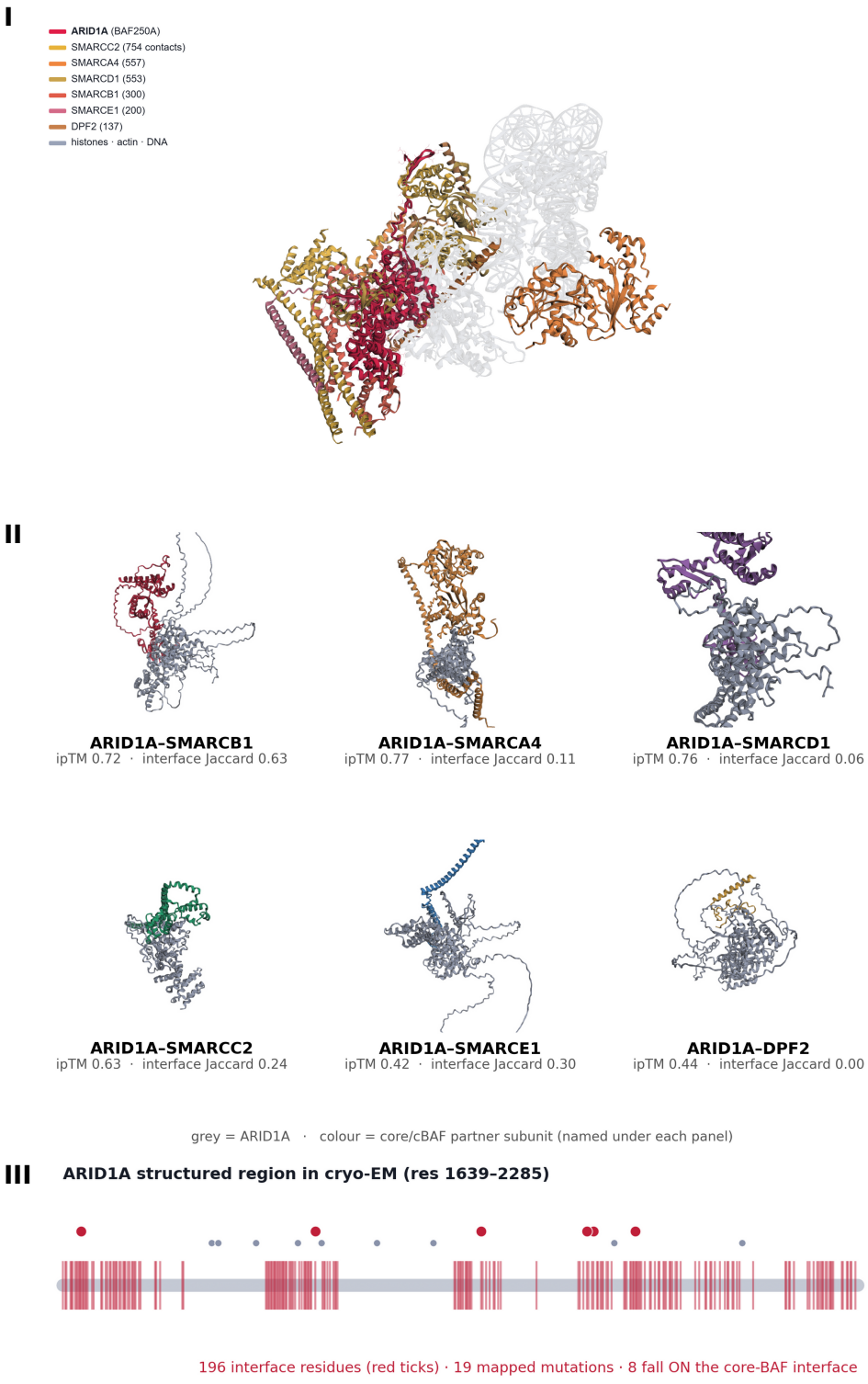


Figure 4: The ARID1A–core-BAF interface: experimental, predicted, and mutated. (**I**) ARID1A within the assembled human cBAF complex (PDB 6LTJ, 3.70 Å) [29], ARID1A in red, each contacting subunit coloured and annotated with its heavy-atom contact count (< 4.5 Å); histones, actin and DNA grey. ARID1A contacts all six core/cBAF subunits at 2.20–2.47 Å, while ACTL6A and ACTB lie 13.0 and 32.0 Å away. (**II**) The same six complexes predicted by AlphaFold-Multimer [33–35] from sequence alone, with self-reported ipTM and Jaccard overlap against the cryo-EM interface. The two highest-confidence models reproduce it worst; the best reproduction ranks only third by ipTM. (**III**) The ARID1A region resolved in 6LTJ (residues 1639–2285): red ticks are the 196 interface residues, circles the 19 NSCLC mutations in this region, eight of them (filled) on the interface.

up to -0.089. The separation is cleaner in lung than pan-cancer, where ACTL6A costs -0.007. SMARCD1 is an apparent exception — despite the second-largest interface its removal improves the pan-cancer score, matching its null co-dependency ($\rho = -0.083$ lung, -0.025 pan-cancer). Separating the paralogues resolves it: the dependency at that position runs through SMARCD2, which no cBAF structure resolves ($\rho = +0.193$, $P = 2.9 \times 10^{-9}$ pan-cancer, tenth of 17,644 genes; $+0.184$, $P = 0.038$ in the smaller lung panel). The structure localises the position, not the gene occupying it.

The refinement generalises to a second complex. Repeating the procedure on PRC2 with the cryo-EM structure of the human PRC2–AEBP2–JARID2 complex [36, 37] and two anchors (Figure 5d), both converge on the catalytic lobe EZH2–EED–SUZ12, with the histone-chaperone subunits RBBP4 and RBBP7 removed: EZH2 improves from 0.607 to 0.746 and SUZ12 from 0.581 to 0.723. RBBP4/RBBP7 are shared with NuRD, CAF-1 and SIN3, so their dependency profiles are driven by complexes other than PRC2.

How far does structural contact go as a decision rule? For two of the three anchors it is exactly right: for ARID1A the contact-restricted core is the best-performing definition, and for EZH2 the contact rule selects EED + SUZ12, which an exhaustive subset search confirms is the optimum. For the third it is not: RBBP4 makes a large, unambiguous contact with SUZ12 (685 heavy-atom pairs across 61 interface residues) and would be retained, yet removing it improves the module by a further $+0.092$. Membership in other complexes is the obvious explanation, and it does not survive testing (Discussion).

Operational recommendation. Structural contact is a good but imperfect prior — correct for two of three anchors — and the criterion should be whether the dropped member replicates on resampling, not the size of the gain, for the reason given in the Discussion. The refinement is substantial and consistent: $+0.109$ (SWI/SNF, lung), $+0.138$ (PRC2, EZH2) and $+0.142$ (PRC2, SUZ12) over the curated definition.

The reproducibility criterion generalises beyond the two examples. We replaced the initial asymmetric survey with a fully nested, symmetric null design (Supplementary Figure S8a). For every split, both the anchor and the member to drop were selected only in the discovery half and carried unchanged into the held-out half; each curated complex was compared with two size-matched random modules under the identical procedure. Of 277 eligible human complexes, 266 yielded a paired comparison over 30 splits. Median held-out drop-identity concordance was 0.60 for curated complexes and 0.333 for matched null modules; the paired median difference was 0.25 (95% bootstrap interval 0.175–0.308; one-sided paired Wilcoxon $P = 1.03 \times 10^{-19}$; Cliff's $\delta = 0.425$). The held-out gain was positive in a median 0.933 of splits for curated complexes and 0.767 for null modules. Thus the transferable result is a nested reproducibility criterion for module refinement, not a universal rule that non-contacting members should be removed.

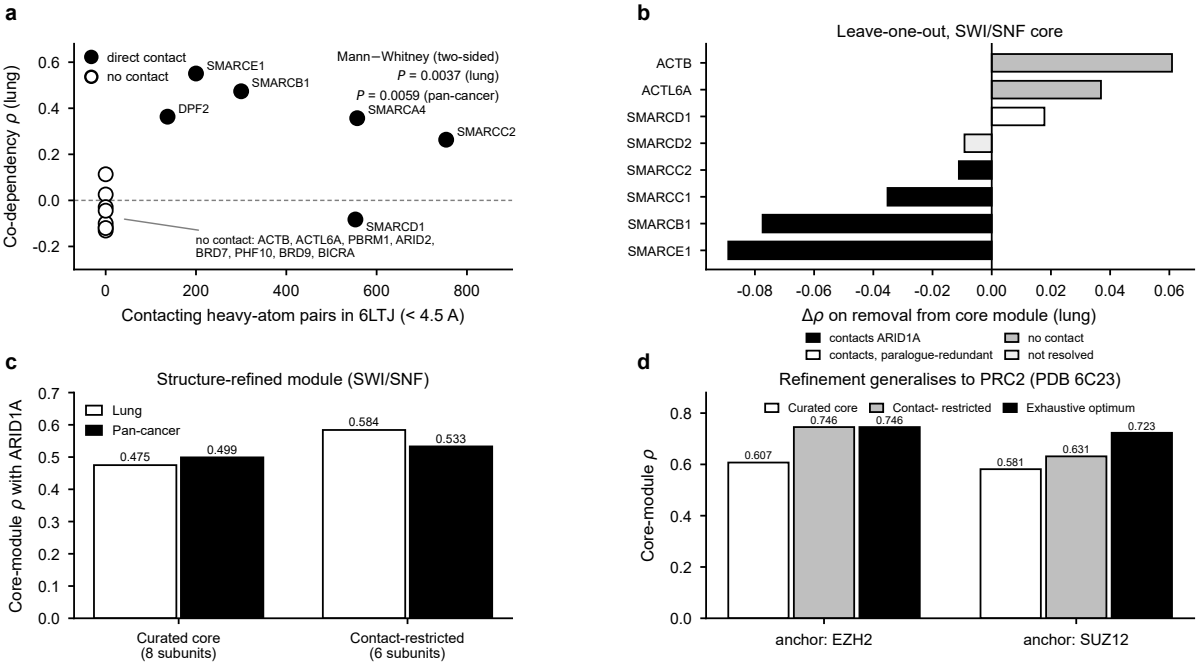


Figure 5: The experimental structure explains and refines the genetic result. (**a**) Contacting heavy-atom pairs in PDB 6LTJ against co-dependency ρ; filled points contact ARID1A directly. The inferential test compares contacting with non-contacting subunits; contact count within the contacting group does not show a positive dose–response. SMARCC1 is not plotted because 6LTJ does not resolve it; its contact is established in 9A0K. (**b**) Leave-one-out change in the core-module correlation in lung, over the same eight subunits used in (**c**), ordered by contact status; SMARCD2 is resolved in no cBAF structure and so has no contact call. (**c**) Curated versus contact-restricted core module in SWI/SNF. (**d**) The same refinement in PRC2 (PDB 6C23) for two anchors, against the exhaustive subset optimum.

2.4 The continuous co-dependency pattern varies with ARID1A expression

If co-dependency reflects shared complex state, its magnitude may vary with the abundance of the anchor. Across *ARID1A* expression quintiles the core-module dependency changes strictly monotonically (-0.706 to -0.510 from lowest to highest), as does the cBAF module, while PBAF (≈ -0.19) and ncBAF ($\approx +0.01$) are flat. The trend is confined to the modules implicated by the co-dependency analysis; partial correlation preserves the specificity (pan-cancer core 0.411, $P = 3.6 \times 10^{-49}$; cBAF 0.218, $P = 4.1 \times 10^{-14}$). Because expression is observational and does not identify protein-null models, this gradient is a consistency analysis rather than evidence of loss-induced sensitivity.

2.5 Generalisation across tissues and independent replication

Across 930 de-duplicated lines from 19 tissues, computed within each tissue, SMARCB1 and SMARCC1 were positive in 19/19 tissues, DPF2 in 18/19, ARID1B 17/19 and SMARCA4 16/19, three to thirteen of them passing global Benjamini–Hochberg correction [38]; PBRM1, ARID2 and SMARCA2 were each positive in 10/19 and negative in 9/19 with *zero* significant tissues (Figure 6a).

Recomputation in a wholly different dataset and pipeline — the DepMap 24Q4 Chronos gene-effect matrix, 1,178 lines [39] — reproduced the pattern (Figure 6b): core/cBAF positive, PBAF/ncBAF/alternative ATPase null, so the finding is not an artefact of one processing pipeline.

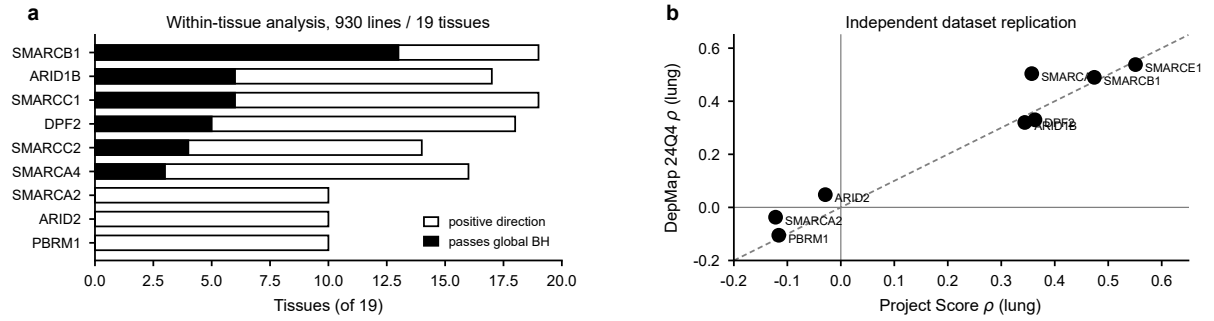


Figure 6: Pan-cancer generalisation and independent replication. (**a**) Within-tissue co-dependency across 19 tissues; open bars count tissues with a positive direction, filled bars those passing global BH correction. (**b**) Project Score versus DepMap 24Q4 co-dependency in lung; the dashed line is the identity.

2.6 Genotype-conditional tests bound the co-dependency interpretation

We next asked the distinct question of whether ARID1A-mutant cell lines are more dependent on individual BAF subunits than wild-type lines. Across five tissues with at least six mutant models, ARID1B had the expected direction in all five, ARID2 and BRD9 in four, and the remaining subunits in one to three. None of the 65 tissue–subunit comparisons survived global Benjamini–Hochberg correction. In lung ($n = 6$ mutant, $n = 122$ wild type), PBRM1 and ARID2 were nominally more essential in mutant lines ($P = 0.0217$ and 0.0248), but both had $q = 0.403$; the principal contacting subunits were non-significant. Thus the continuous co-dependency pattern is robust, whereas this dataset does not establish an ARID1A-loss-conditioned vulnerability.

We therefore repeated the analysis using ARID1A protein abundance as the primary state definition and mRNA abundance as a larger-sample sensitivity definition. Gene effects were standardized within lineage before pooling, and the bottom 5%, 10% and 15% of ARID1A abundance were compared with the remainder. The protein analysis included 535 usable lines (25, 52 and 80 in the low groups); no individual subunit or prespecified BAF module passed Benjamini–Hochberg correction at any threshold. At the protein bottom 10%, ARID1B was directionally more dependent (mean within-lineage z difference -0.277, 95% bootstrap interval -0.608 to $+0.048$; $q = 0.187$), whereas the ARID1A-contacting module was directionally less dependent ($+0.152$, 0.013–0.292; $q = 0.187$). The mRNA analysis gave the same corrected-null conclusion. As an assay-dynamic-range control, the identical pooled design strongly detected the established SMARCA4-low–SMARCA2 dependency: at the mRNA bottom 10%, $Deltaz = -1.012$ (-1.355 to -0.689; $q = 3.05 \times 10^{-9}$). The negative ARID1A result is therefore not explained by an intrinsically insensitive pooled pipeline (Supplementary Figure S8d), but protein-defined low abundance is still not equivalent to engineered biallelic loss.

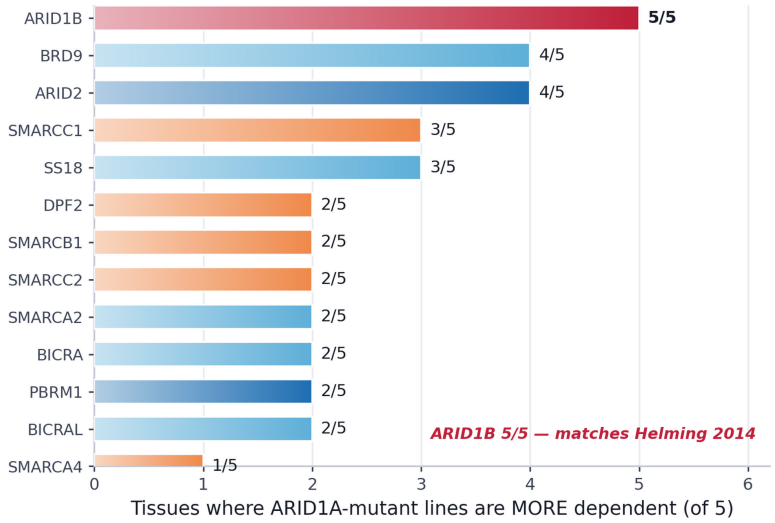


Figure 7: Genotype-conditional dependency is directional but underpowered. For each BAF subunit, the number of tissues (of five with adequate sample size) in which ARID1A-mutant cell lines are more dependent than wild-type lines. ARID1B has the expected direction in 5/5 tissues, but no tissue–subunit comparison survives global multiple-testing correction. This analysis measures a different axis from the continuous co-dependency gradient and is reported as a boundary on interpretation, not as evidence of synthetic lethality.

2.7 A self-correlation artefact must be corrected

Module scoring correlates an anchor's dependency against a module mean. If the anchor is itself a member, $1/k$ of the module score *is* the anchor. Across three complexes and 48 anchor–module pairs all 48 were inflated by more than 0.10, with inflation scaling inversely with module size (Spearman $\rho = -0.597$, $P = 7.3 \times 10^{-6}$; mean $+0.506$ at $k = 2$, $+0.358$ at $k = 3$, $+0.197$ at $k = 9$). At small k the consequence is a reversed conclusion, not a shifted estimate: STAG2 appears strongly co-dependent with its own cohesin module ($+0.839$) when the anchor-excluded value is -0.045, not significantly different from zero. All analyses here exclude the anchor from every module.

That correlating a variable with a mean containing it inflates the correlation is elementary; the contribution here is the empirical magnitude and the demonstration that at small k it reverses conclusions. We have not surveyed how often the anchor is left in, and make no claim about prevalence.

Benchmark with the correction applied. The shared core module is common to all sub-complexes, so a high core score is expected for *every* anchor; the testable hypothesis is one of discrimination. Scored over fourteen anchors of known mutually exclusive membership across SWI/SNF and PRC2, all fourteen are assigned correctly, competing modules near zero or negative throughout, and SMARCA2 returns null on all four modules ($|\rho| \leq 0.042$) — correct for an alternative ATPase belonging to no sub-complex. Since co-essentiality recovering complex membership is established [14, 15], this is a positive control, not a novel finding.

Null controls. Three hundred random genes outside the complex give a best exclusive-module correlation of +0.028 (median; 95th percentile +0.095); only 4.0% pass a $\rho > 0.1$ threshold and only 0.3% exceed the weakest true anchor (Mann–Whitney $P = 1.7 \times 10^{-7}$). Decoy anchors from unrelated complexes give a median of +0.091, but not all are inert: MED12 scores +0.241 on ncBAF and the three PRC2 core subunits lean toward ncBAF (+0.138 to +0.181). Thresholds in the 0.1–0.25 range therefore admit false positives from functionally related complexes, and assignments should carry their margin over the best competitor.

2.8 Computational structural methods cross-validate one interface, and one confidence metric fails

Blind rigid-body docking [40] with 200 swarms (6,000 poses per partner) located the correct interface for all four partners tested (best Jaccard 0.325–0.500; Figure 8a). Top-ranked poses perform poorly because the scoring function ranks large scaffold interfaces badly, so we treat this only as evidence that the interface is findable.

AlphaFold-Multimer [33–35] run from sequence alone accurately recovered the ARID1A–SMARCB1 interface (87% recall, Jaccard 0.628). But across the six partners **ipTM and interface recovery are unrelated** (Spearman $\rho = -0.14$, $P = 0.79$; Figure 8b), and the two highest-confidence predictions are among the worst reproductions, consistent with benchmarking showing that multimer confidence metrics degrade for large and asymmetric assemblies [41, 42] and with the incomplete structural coverage of the human interactome [43]. ipTM should not be used as a reliability filter when validating interfaces within large assemblies. The six models are shown in Figure 4II.

The interface is also where a substantial share of the mutations fall: of the 73 ARID1A mutations in the NSCLC cohort, 19 lie within the region resolved in 6LTJ (residues 1639–2285) and eight of those land on the 196-residue core-BAF contact surface (Figure 4III). Because most ARID1A mutations are truncating (Figure 2b) and remove the region entirely, this is not evidence for a separable interface-disrupting mutation class; it records only that the contact surface lies within the mutated territory.

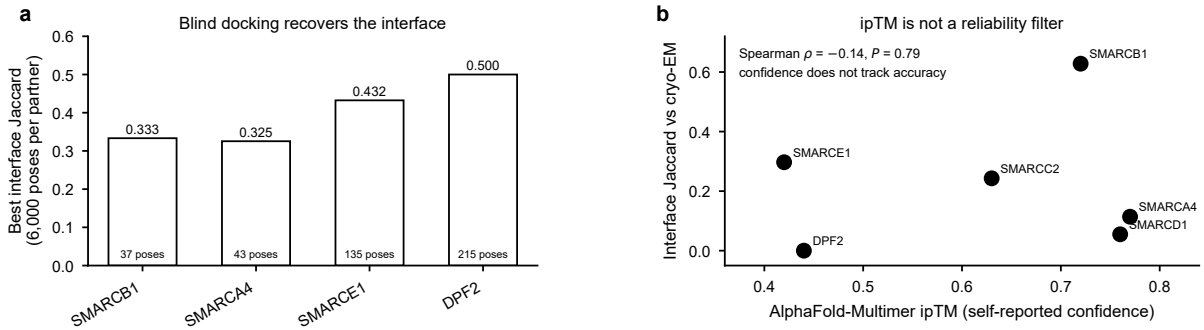


Figure 8: Computational structural cross-validation. (**a**) Blind docking interface recovery; the best Jaccard over all 6,000 poses per partner, with the number of near-native poses ($J \geq 0.2$) annotated. (**b**) AlphaFold-Multimer interface reproduction against self-reported ipTM for the six contacting partners.

2.9 Alternative explanations excluded

The complex the dependency targets is expressed in real tumours. In 206,826 Harmony-integrated cells from a NSCLC single-cell atlas [44, 45], ARID1A and the residual core/cBAF subunits are expressed in the malignant epithelial compartment (Supplementary Figure S3). To respect the nesting of cells within specimens, we aggregated the 35,600 epithelial cells to 48 sample-level observations. Mean ARID1A expression was modestly associated with the mean core/cBAF score (Spearman $\rho = 0.416$, $P = 0.0033$); samples below versus above the median differed by only 0.0147 score units (medians -0.0179 versus -0.0032; exact Mann–Whitney $P = 0.0257$). The small sample-level effect shows that the genes are detectable in tumour epithelium but does not establish that low ARID1A expression is functional loss or that an intact residual complex persists. Human Protein Atlas immunohistochemistry agrees at the protein level, reporting moderate-to-strong nuclear ARID1A and strong nuclear SMARCB1 and SMARCA4 staining across malignant tissue including lung [46].

Transcript-level data cannot resolve complex specificity. *In silico* deletion of ARID1A with a single-cell foundation model [47] in 7,270 malignant epithelial cells [44] returned null for BAF subunit ranks ($P = 0.3$–0.6) in two independent runs; the undownsampled run increased detection counts three- to four-fold and remained null. An architecturally different foundation model agrees from the opposite direction: in scGPT's representation [48] the SWI/SNF subunits sit close to ARID1A regardless of sub-complex, with PBRM1 its nearest SWI/SNF neighbour despite a genetic PBAF co-dependency of essentially zero (Supplementary Figure S3). Bulk mRNA and cell-line proteomic correlations [49] likewise place ARID1A in positive correlation with *all thirteen* BAF family genes, including the mutually exclusive PBRM1 and SMARCA2 (Figure 9a), so a conclusion about complex specificity drawn from the sign of an expression correlation would be demonstrably wrong at the genetic level.

The specificity is present at the protein layer but not the transcript layer. Sign is not the informative feature; magnitude is. On the 677 cell lines carrying both measurements, the same lines used for both, whether ARID1A contacts a subunit is invisible in the transcriptome (median ρ 0.571 contacting versus 0.432 not; AUC 0.57, $P = 0.69$) and plainly visible in the proteome (0.598

versus 0.207; AUC 0.96, $P = 0.0012$), where the correlation also tracks interface size ($\rho = +0.65$, $P = 0.0086$, against $+0.18$, $P = 0.53$ for transcripts). The comparison is confounded: protein correlation tracks detection rate and abundance ($\rho = +0.82$ for both) and contacting subunits are the better-measured ones. Restricted to the nine subunits detected in over 90% of lines the protein-level difference persists (0.605 versus 0.385, $P = 0.048$) while the transcript-level difference stays absent and reverses in sign, so we report the contrast as supported but abundance-limited.

An exploratory Mendelian-randomization analysis did not provide evidence for a germline ARID1A-expression effect on lung-cancer risk (Figure 9b).

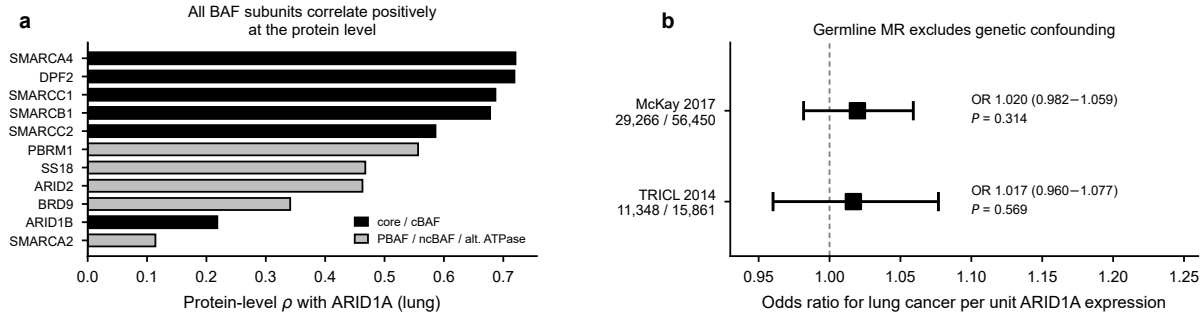


Figure 9: Expression-layer comparison and exploratory germline analysis. (**a**) Protein-level correlation of ARID1A with BAF family subunits in lung cell lines; all subunits correlate positively regardless of sub-complex membership. (**b**) Two-sample Mendelian randomization [50] using ARID1A *cis*-eQTL instruments [51] against two independent lung cancer GWAS [52]; squares are odds ratios with 95% confidence intervals. Both are null (OR = 1.020, $P = 0.314$; OR = 1.017, $P = 0.569$). Only one non-lung instrument was available: ARID1A carries seven independent *cis*-eQTLs across 49 GTEx tissues [53] and none in lung, as expected for a haploinsufficient tumour suppressor under purifying selection [54]. The null estimates are therefore weakly informative and cannot exclude genetic confounding.

2.10 Pharmacological data do not establish genotype selectivity

Thirteen GDSC compounds targeting PARP, ATR, ATM, EZH2 or SWI/SNF were compared between ARID1A-mutant and wild-type models in lung and pan-cancer strata. No comparison survived within-stratum Benjamini–Hochberg correction ($q \geq 0.27$). PFI-3 and I-BRD9 were among the least active compounds in the lung panel (mean AUC 0.977 and 0.956), leaving little dynamic range for a genotype interaction. These data are therefore uninformative rather than evidence against selectivity. Likewise, genotype-conditional CRISPR effects for SMARCA4 and SMARCA2 did not survive correction. The present study consequently makes no therapeutic-efficacy claim: co-dependency prioritizes functional relationships, not drug targets, and requires protein-loss-defined or isogenic validation before pharmacological translation.

As a bounded structural-plausibility check, redocking PFI-3 [55] into its crystallographic SMARCA2/4 bromodomain pocket (PDB 5DKC) with AutoDock Vina [56] reproduced the deposited pose at 0.48 Å RMSD and gave a score of $-8.4\,\mathrm{kcal\,mol^{-1}}$ (Figure 10A,B; Supplementary Figure S4). Blind docking of the clinical-stage dual SMARCA2/4 ATPase inhibitor FHD-286 [57] into PDB 7VDT gave $-9.17\,\mathrm{kcal\,mol^{-1}}$ (Figure 10A; Supplementary Figure S6). No public co-crystal validates that pose, however, and neither docking score is a measured affinity or evidence of ARID1A-genotype selectivity. As a secondary pose-persistence check, one explicit-solvent

trajectory per complex was analysed (Supplementary Figure S5). Pocket-fitted ligand RMSD was 1.75 ± 0.88 Å for PFI-3 over 30 ns and 2.21 ± 0.21 Å for FHD-286 over 20 ns; the corresponding binding-site backbone RMSD values were 0.54 ± 0.13 and 1.33 ± 0.16 Å. Single-trajectory MM/GBSA endpoint estimates were -27.93 ± 2.33 and -33.45 ± 4.29 kcal mol^{-1}, respectively. These framewise summaries show bounded behaviour within the sampled trajectories only. They do not validate the blind-docked FHD-286 pose, estimate absolute affinity or establish genotype selectivity, and remain outside the primary evidentiary chain.

Not every tractability argument rests on a predicted pose. The ACBI1 ternary complex has been resolved experimentally by X-ray diffraction (PDB 7S4E, 2.25 Å), directly showing engagement of the human SMARCA2 bromodomain by ACBI1 together with VHL–ElonginC–ElonginB [58]. We did not redock or resimulate that deposited complex. It provides a structural precedent for ligandable SMARCA2 engagement and targeted degradation, but it does not establish that ARID1A-deficient cells are selectively sensitive to ACBI1 or to any other SMARCA2/4-directed agent.

An archived AutoDock Vina screen of a 6,392-entry project compound library was also recovered and audited (Supplementary Figure S7). Within the project's -13 to -5 kcal mol^{-1} artefact-exclusion window, 5,311 bromodomain and 5,493 ATPase scores remained. YM-750, arcyriaflavin A and BINA led the bromodomain ranking; MK-3207, BT-11 and SR-1664 led the ATPase ranking. The run was resumed at lower exhaustiveness and the exact library source/version metadata were not retained, so the result is reported only as a ranking sensitivity analysis. These scores are neither binding affinities nor evidence that the compounds are repurposing candidates.

3 Discussion

Five methodologically distinct analyses of one CRISPR dependency matrix converge on core-shared and cBAF-associated subunits. The pattern survives pan-essentiality and lineage controls, is reproduced in a second processing resource and across 19 tissues, and aligns with a binary contact partition in an experimental structure. Together these results map co-essential organization of the ARID1A-associated BAF module. They do not show that ARID1A loss creates the pattern, because mutation-, protein- and mRNA-defined conditional analyses are non-significant after correction even though the pooled design detects a known paralogue-dependency positive control.

Curated module definitions can be tested empirically. The 266-complex nested survey shifts the methodological emphasis from maximizing an in-sample correlation to reproducing the identity of a dilutive member in held-out data. Applying the same discovery-half anchor and drop selection to curated and matched-null modules removes the main selection asymmetry of the initial survey; the paired concordance advantage remains large. This criterion is useful beyond SWI/SNF, while remaining a statistical refinement rather than proof that the nominated member is biologically dispensable.

Structure is a prior, not a rule. We expected structural contact to supply the decision criterion. In SWI/SNF the prior happens to be correct, because the two dilutive subunits are

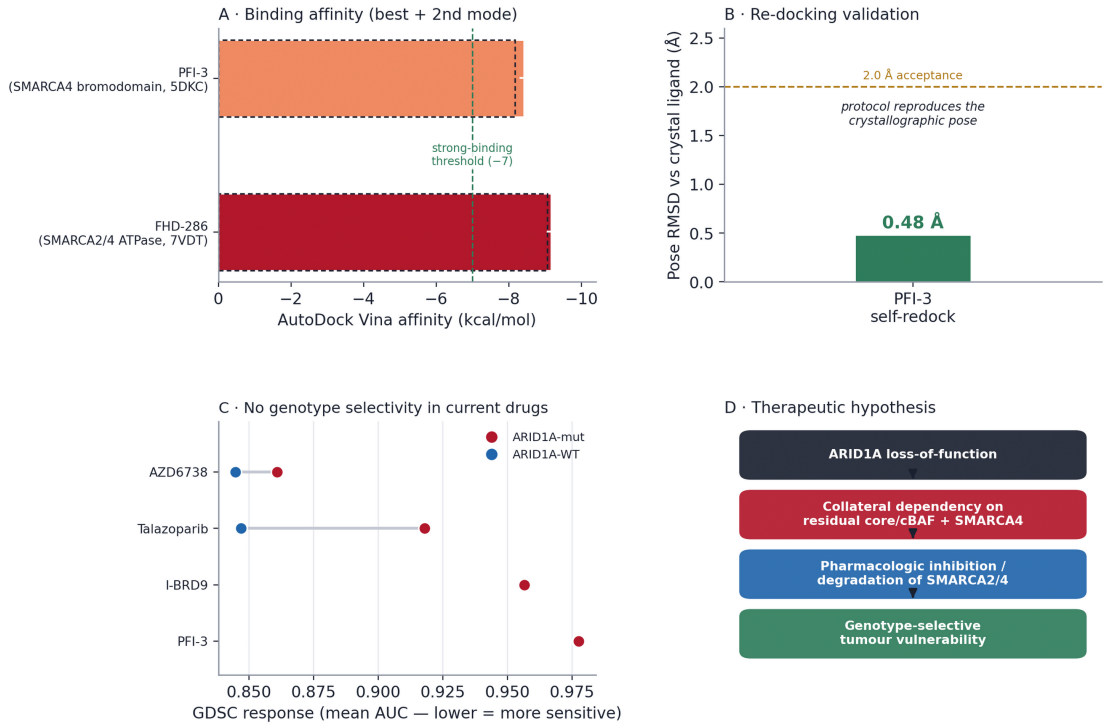


Figure 10: Exploratory target-engagement plausibility does not establish ARID1A-genotype selectivity. (**A**) AutoDock Vina scores for PFI-3 in the SMARCA2/4 bromodomain (PDB 5DKC) and FHD-286 in the SMARCA4 ATPase nucleotide pocket (PDB 7VDT); dashed outlines mark the second-ranked mode. (**B**) Positive-control redocking reproduces the PFI-3 crystallographic pose at 0.48 Å RMSD. (**C**) Representative GDSC responses stratified by ARID1A genotype. None of the thirteen tested compounds survived within-stratum Benjamini–Hochberg correction; PFI-3 and I-BRD9 were near the inactive end of the lung panel, limiting the ability to detect an interaction. (**D**) Bounded therapeutic hypothesis linking ARID1A loss, collateral BAF dependence and pharmacological inhibition. Panels A, B and D are exploratory: docking scores rank modelled poses rather than measuring binding affinity, and the diagram is a hypothesis rather than an efficacy claim.

also the two non-contacting ones; PRC2 dissociates the properties, since RBBP4 contacts SUZ12 substantially yet dilutes. Subunit promiscuity is the obvious candidate, and it fails: the number of curated human complexes containing each subunit [59] — an external count, not one derived from the dependency data — does not predict the leave-one-out delta (Spearman $\rho = +0.21$, $P = 0.63$ within SWI/SNF), and dilutive and load-bearing subunits have indistinguishable counts ($P = 0.91$). Curation depth confounds such counts, so this bounds rather than closes the question. We say so explicitly because the purely structural rule would be the more attractive claim and is not what our data support.

Two layers that do not substitute for one another. Every BAF family gene correlates positively with ARID1A in both mRNA and protein, mutually exclusive subunits included, so the sign of an expression correlation carries no information about complex identity. What separates the layers is that the protein-level magnitude reproduces the structural partition while the transcript-level magnitude does not — an orthogonal measurement type, on the same cell lines, recovering what the CRISPR screens give. Transcriptional co-regulation is a poor proxy for complex membership; protein abundance is a usable one, within the limits of what mass spectrometry detects.

Relationship to existing paradigms. ARID1B shows a *positive* continuous co-dependency with ARID1A, superficially opposite to synthetic lethality [10], and is robust across pipelines. In the separate genotype-stratified analysis ARID1B had the expected direction in all five tissues, and the protein- and mRNA-defined pooled analyses also pointed toward greater ARID1B dependency, but no comparison survived multiplicity correction. The same pooled pipeline recovered the SMARCA4-low–SMARCA2 positive control, so the corrected-null ARID1A result is biologically informative rather than a generic lack of assay sensitivity. It is compatible with the established ARID1A–ARID1B relationship but does not independently reproduce it. More broadly, continuous co-dependency across heterogeneous lines and conditional essentiality after complete loss are different estimands. Protein-defined null models or isogenic perturbations are needed before translating the former into a synthetic-lethal claim.

Relationship to residual-complex compensation. In dedifferentiated endometrial carcinoma, ARID1A/ARID1B deficiency has been shown to reduce cBAF while increasing the biochemical abundance and chromatin activity of ncBAF and PBAF, accompanied by sensitivity to disruption of residual complexes [13]. Our continuous co-dependency map points in a different direction: core/cBAF-associated subunits are positive, whereas the defining PBAF/ncBAF subunits are null or negative. The two results need not conflict. The endometrial study measures a loss-conditioned change in complex abundance and function in a defined tumour state; we measure baseline co-essential covariance across heterogeneous cell lines. The apparent contrast is therefore a testable context and estimand difference, not evidence that either layer substitutes for the other.

Limitations of the study

The study is entirely computational and correlative. Positive co-dependency identifies cell lines in which two genes tend to be essential together; it does not establish that loss of one gene creates dependence on the other. The mutation-defined genotype tests were underpowered and non-significant after correction. Protein- and mRNA-defined low-abundance analyses used more lines and an internally validated pooled design but likewise yielded no corrected module dependency; abundance tails do not guarantee biallelic loss or absence of residual protein. Isogenic perturbations and rescue experiments remain the necessary next tests of loss-conditioned vulnerability.

The structural contact analysis supplies an orthogonal physical rationale rather than a mechanism. The full 6LTJ model has a poor global wwPDB Q-score relative to same-resolution entries; the binary interface is supported by the higher-quality 6LTH replicate, distance-threshold sensitivity and residue-Q filtering, but fine atom-level interpretation of 6LTJ is not warranted. The analysis uses static wild-type assemblies and therefore cannot show the composition or conformation of a residual complex after ARID1A loss. Contact status separates the co-dependency groups, but contact count does not show a positive dose–response among resolved contacts. Co-immunoprecipitation or native-gel assays would be needed to test complex integrity directly.

Module refinement is characterised mechanistically on two complexes and three anchors and surveyed statistically across 266; the symmetric nested survey establishes held-out reproducibility of the nominated member relative to a matched null, not that dropping it is biologically correct, and the mechanism is unresolved. Two subunits are placed by an integrative model rather than refined density: SMARCC1, which no experimental cBAF structure resolves, and SMARCD2, whose position is inferred from its paralogue. Computational structural methods validate only the ARID1A–SMARCB1 interface effectively; the 14/14 discrimination benchmark is a positive control rather than a novel finding, and decoy controls are not fully inert. The five discovery analyses share one dependency matrix and are methodologically distinct rather than biologically independent. The single-cell result is sample-aggregated but remains an expression analysis, and the transcript- and foundation-model negatives are unexplained mechanistically. The molecular-dynamics analyses comprise one short trajectory per complex; their framewise variation is not uncertainty across independent simulations. Moreover, MM/GBSA omits conformational entropy and FHD-286 starts from a blind-docked rather than crystallographic pose. These analyses test persistence of the modelled poses over the sampled windows, not pose correctness, absolute binding affinity or therapeutic efficacy. The recovered virtual screen combines an initial run at Vina exhaustiveness 8 with a resumed run at exhaustiveness 4, and its exact compound-library source/version metadata were not archived. Consequently its ranks are exploratory, are not directly comparable to a uniform-exhaustiveness screen and cannot support repurposing claims.

Methods

Data sources. TCGA-LUAD ($n = 514$) and TCGA-LUSC ($n = 485$) [18]; the Sanger Project Score CRISPR screen, de-duplicated by model identifier (lung $n = 128$; pan-cancer 930 lines, 19 tissues) [24, 25]; the DepMap 24Q4 Chronos matrix (1,178 lines) [39]; PDB 6LTJ, 6LTH, 6C23

and 9A0K; GEO GSE131907 [44]; OpenGWAS accessions for the eQTL exposure [51] and lung cancer outcomes [52].

Mutation landscape and differential expression. maftools [60] on the clinically matched cohort denominator; other MAF sources give slightly different denominators (6.1–6.8%), so the cohort definition must be stated for reproduction. DESeq2 [23] on TCGA-LUAD by *ARID1A* expression tertile, significance at adjusted $P < 0.05$ and $|\log_2 \mathrm{FC}| > 1$.

Independent mutation-frequency replication. Two non-TCGA lung adenocarcinoma cohorts queried through the cBioPortal REST API [19, 20]: `luad_mskimpact_2021` ($n = 186$) [21] and `luad_mskcc_2023_met_organotropism` ($n = 2{,}653$) [22]. A sample was called mutant if it carried at least one non-synonymous *ARID1A* variant; frequency is mutant over all sequenced samples. Both are targeted panels covering ARID1A, ARID1B, SMARCA4, SMARCB1, SMARCD1, PBRM1 and ARID2 (SMARCE1 and SMARCA2 added in v505), which is why the replication is restricted to mutation frequency.

CRISPR co-dependency. Spearman correlation with global Benjamini–Hochberg correction [38]. **Tissue-level correlations are computed within tissue**; pan-cancer values are pooled, the lineage control being the separate 19-tissue analysis. Module score is the mean gene effect over module members, **with the anchor excluded from every module**. Table 1 uses Project Score; the refinement and leave-one-out analyses use DepMap 24Q4, which is why the curated core's correlation differs between them. Prior-free enrichment via Enrichr [26] on the top-100 co-dependent genes only. Co-essentiality network from 7,940 selectively essential genes with Leiden community detection [27], community identity interpreted only after detection. The archived feature-attribution analysis used XGBoost with SHAP [28]. For reproducible sensitivity analysis, the DepMap 24Q4 matrix was restricted to the 3,000 non-target genes with highest variance, missing values were median-imputed, and prediction was evaluated by five-fold out-of-fold cross-validation (shuffle seed 20260831). XGBoost used 500 trees, maximum depth 3, learning rate 0.03, subsample and column-sample fractions 0.8, and L_2 regularisation 1.0; SHAP values were then computed from a model fitted to all 1,178 lines. Null controls: 300 random non-member genes and 13 decoy anchors from PRC2, cohesin, ribosome, proteasome, RNA polymerase II and Mediator, scored identically.

Structural analysis. Biopython [31] parsing of PDB 6LTJ and 6LTH (SWI/SNF) and 6C23 (PRC2); minimum heavy-atom distance and count of heavy-atom pairs within 4.5 Å between the anchor chain and every other chain; chain identity verified against PDB COMPND records; exact two-sided Mann–Whitney comparison of contacting versus non-contacting groups, with Cliff's delta and a 20,000-resample percentile bootstrap interval for the median difference (seed 20260831); Spearman correlation of contact count with co-dependency was assessed separately among the six contacts resolved in 6LTJ. Leave-one-out and exhaustive-subset module recomputation. In 6C23 subunits are modelled as disjoint fragments rather than separate copies, so per-gene contacts are summed across fragments and a contact required at least three interface residues. The same

criterion placed SMARCC1 using the integrative model 9A0K [32]; built from crosslinking mass spectrometry and electron-microscopy envelopes rather than refined density, it is used only for the binary contact call, and modelled residue coverage was compared subunit by subunit first. Subunit promiscuity was the number of EBI Complex Portal complexes containing that protein [59], tested against the leave-one-out delta by Spearman correlation.

Structure-quality and contact-call sensitivity. Official wwPDB validation XML reports and deposited mmCIF coordinates were parsed for 6LTJ and 6LTH. We extracted resolution, EMDB accession, global and chain Q-scores, same-resolution percentiles, atom inclusion, clashscore, Ramachandran outliers and rotamer outliers. Heavy-atom contacts were recomputed at 4.0, 4.5 and 5.0 Å. To test sensitivity to locally weak model support, residues in the bottom decile of wwPDB residue Q-score were removed separately within each analysed chain before contact calls were repeated. Interface versus non-interface ARID1A residue Q-scores were compared by two-sided Mann–Whitney, and shared ARID1A residues were compared between deposits by paired Wilcoxon. Structural genetic-group sensitivity used exact two-sided Mann–Whitney, Cliff's delta and 20,000-resample percentile bootstrap intervals (seed 20260901) for the original, experimental-contact-only and physically-present-subunit definitions, plus leave-one-gene-out analyses.

Protein versus transcript layer. Sanger cell-line proteomics [49] and CCLE expression were matched through Cell Model Passports identifiers, giving 677 lines with both. Per subunit, the ARID1A-partner Spearman correlation was computed at the protein and the transcript level on the identical lines, so sample composition cannot generate a difference between layers, and the contacting and non-contacting groups compared within each layer by two-sided Mann–Whitney. Because protein correlation is not independent of how well a protein is measured, detection rate and abundance were tested as confounders and the comparison repeated on subunits detected in over 90% of lines.

Generalisation survey. Complex Portal human complexes containing 6–21 DepMap-covered subunits were eligible (277). In each of 30 random half-splits, the highest-correlating anchor and the member whose removal most improved the anchor-excluded module score were selected using only the discovery half, then carried unchanged into the held-out half. Two random gene modules matched to each complex's size were processed identically, including discovery of their anchor and dropped member. A complex contributed to the paired analysis when the real and matched-null modules had valid results (266 complexes). Primary outcome was held-out drop-identity concordance; secondary outcome was the fraction of splits with positive held-out gain. Real-minus-null differences were summarized at the complex level, with a 20,000-resample paired bootstrap interval, one-sided paired Wilcoxon test and Cliff's delta (seed 20260901).

Interface prediction. AlphaFold2-Multimer v3 via ColabFold [33–35] on the ARID1A domain (UniProt 1639–2285) with partner sequences; interface compared on the ARID1A side at 4.5 Å. Blind rigid-body docking with LightDock [40] used 200 swarms and 6,000 poses per partner as an

interface-recovery sensitivity analysis.

Exploratory small-molecule docking. AutoDock Vina 1.2 [56] was used for a positive-control redocking of PFI-3 into the SMARCA2/4 bromodomain co-crystal structure (PDB 5DKC) and for blind docking of FHD-286 into the SMARCA4 ATPase nucleotide pocket (PDB 7VDT). The PFI-3 control was evaluated by heavy-atom RMSD from the deposited ligand pose. Vina scores were treated only as within-protocol pose scores; no conversion to binding affinity and no genotype-selective activity inference was made.

Exploratory virtual screen. An archived 6,392-entry project compound library was converted from SMILES to PDBQT with Open Babel using three-dimensional coordinate generation and protonation at pH 7.4. One Vina mode was retained per compound and target. For the 5DKC bromodomain, the search box was centred at $(12.52, 33.77, 60.45)$ Å with dimensions $22 \times 22 \times 22$ Å; for the 7VDT ATPase pocket, the centre was $(126.04, 201.63, 244.04)$ Å with dimensions $24 \times 24 \times 24$ Å. The archived result combines the initial run at exhaustiveness 8 with a recovery run at exhaustiveness 4. The audit retained the project's -13 to -5 kcal mol^{-1} display window and recorded missing and out-of-window values rather than treating them as hits. Because the exact library source/version metadata were not retained and exhaustiveness was not uniform, ranks are presented only as hypothesis-generating within-target scores.

Exploratory molecular dynamics and MM/GBSA. The PFI-3–SMARCA2 bromodomain and FHD-286–SMARCA4 ATPase systems were constructed with AmberTools using ff14SB protein parameters [61], GAFF2 ligand parameters and TIP3P water, then propagated in OpenMM [62]. Simulations used particle-mesh Ewald electrostatics, a 1.0-nm non-bonded cutoff, constrained bonds to hydrogen, hydrogen-mass repartitioning and a 4-fs step. After minimisation and 300 ps of position-restrained NPT equilibration, one unrestrained production trajectory was generated per complex at 300 K and 1 atm (PFI-3, 30 ns; FHD-286, 20 ns), saving coordinates every 10 ps. Ligand RMSD was calculated after fitting binding-site residues, and binding-site backbone RMSD was calculated separately. Single-trajectory MM/GBSA [63] used the GB model igb=2 with 0.15 M salt, sampling from frame 100 at stride 10 and omitting conformational entropy. Values are descriptive endpoint estimates from single trajectories, not replicate-based affinity estimates.

Protein-level expression in tumour tissue. Immunohistochemistry annotations for ARID1A, SMARCB1 and SMARCA4 were read from the Human Protein Atlas cancer-pathology pages [46] — the resource's own qualitative staining calls, used only as corroboration that the complex is present at protein level in malignant tissue.

Single-cell analysis. GSE131907 [44] integrated across samples with Harmony [45]; 206,826 cells after quality control. The core/cBAF module score is the mean scaled expression of the eight shared-core subunits per cell. For inference, epithelial-cell ARID1A expression and module score were averaged within each of 48 samples, then assessed by Spearman correlation and an

exact Mann–Whitney comparison after a sample-level median split. Cell-level distributions are visualized descriptively only.

Genotype-stratified drug response. GDSC area-under-curve values for thirteen compounds — PARP (olaparib, talazoparib, niraparib, rucaparib, veliparib), ATR (AZD6738, AZ20, VE-822), ATM (KU-55933, KU-60019), EZH2 (GSK343) and the two SWI/SNF-directed probes PFI-3 and I-BRD9 — were compared between ARID1A-mutant and wild-type cell lines by Mann–Whitney test, separately in lung-only and pan-cancer strata. The lung stratum is a subset of the pan-cancer stratum for all thirteen compounds, so the two sets of tests are not independent and Benjamini–Hochberg correction was applied within each stratum rather than across the pooled 26 comparisons. Minimum detectable differences were computed post hoc at $\alpha = 0.05$ and 80% power over a plausible range of within-group standard deviations (0.06–0.10 AUC). To establish whether the assay had dynamic range, each compound's mean AUC was ranked against all 286 compounds profiled in the same lung panel.

Genotype-conditional dependency. For each BAF subunit, DepMap gene effect was compared between ARID1A-mutant and wild-type lines within each tissue having at least six mutant lines (five qualified: endometrium, ovary, large intestine, stomach, lung), by Mann–Whitney with global Benjamini–Hochberg correction. Subunits are summarised by the number of tissues in which mutant lines are the more dependent group, a categorical read-out distinct from the continuous gradient used elsewhere.

For abundance-defined sensitivity analyses, Sanger ARID1A protein abundance was the primary anchor and CCLE ARID1A mRNA the secondary anchor. For every tested dependency gene, effects were standardized within OncoTree lineage; lineages with fewer than eight models were excluded, after which lines were pooled. Bottom 5%, 10% and 15% abundance groups were compared with the remainder by two-sided Mann–Whitney, with Benjamini–Hochberg correction within each threshold. Prespecified module scores were per-line means of the standardized gene effects for ARID1A-contacting cBAF, shared core, cBAF-specific, PBAF-specific and ncBAF-specific members; ARID1B and SMARCA2 were also tested individually. Mean differences and 95% percentile bootstrap intervals used 20,000 resamples (seed 20260901). The assay-dynamic-range control applied the identical design to SMARCA4 mRNA abundance and SMARCA2 dependency, with correction across the three thresholds.

Virtual knockout and Mendelian randomization. Geneformer [47] on GSE131907 malignant epithelial cells, run on a 2,000-cell downsample and on all 7,270 cells; scGPT [48] on the same cells, with transcriptional-neighbour percentiles the rank of each SWI/SNF subunit's cosine similarity to ARID1A among all 13,032 genes in the embedding. Two-sample Mendelian randomization with MR-Base [50], Wald ratio.

Author contributions

Zhiyong Tao: Conceptualization; Methodology; Software; Formal analysis; Investigation; Visualization; Writing — original draft. **Juan Xiong:** Methodology; Formal analysis; Validation; Data curation; Investigation; Writing — original draft. **Yiwei Zhang:** Methodology; Software; Formal analysis; Validation; Data curation; Visualization; Writing — original draft. **Chulin Zhong:** Data curation; Formal analysis; Validation; Visualization. **Wenkai Gao:** Data curation; Investigation; Validation; Writing — review & editing. **Guohua Fan:** Conceptualization; Supervision; Project administration; Resources; Writing — review & editing.

Zhiyong Tao, Juan Xiong and Yiwei Zhang contributed equally to this work. All authors read and approved the final manuscript.

Key Points

- ARID1A-contacting subunits show higher co-dependency than non-contacting subunits, but contact count does not predict co-dependency within resolved contacts; structure is a binary prior, not a quantitative rule.
- Curated protein-complex module definitions are sub-optimal for dependency analysis; refining the SWI/SNF core raises its correlation from 0.475 to 0.584 while dropping two of eight genes, and a symmetric nested test across 266 complexes outperforms size-matched null modules in held-out data.
- Module-score co-dependency is inflated unless the anchor is excluded from its own module, which reversed conclusions in two of the modules tested here.
- Five methodologically distinct analyses with pan-essentiality and lineage controls assign 14/14 anchors of known membership to their correct mutually exclusive module across two complexes.
- Mutation-, protein- and mRNA-defined ARID1A-low tests do not survive multiple-testing correction despite detection of a SMARCA4-low–SMARCA2 positive control, so the study maps co-essential organization rather than synthetic lethality.
- Transcript-level data cannot resolve complex specificity, and AlphaFold-Multimer's ipTM does not track interface accuracy within a large assembly.

Funding

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Conflict of interest

The authors declare no competing interests.

Declaration of generative AI use

During the preparation of this work the authors used generative AI assistants (Anthropic Claude and OpenAI Codex) to write and execute analysis code, to generate figures from the resulting data, and to draft and revise manuscript text. All analyses were specified by the authors, all outputs were checked against the underlying data, and the authors reviewed and edited the content and take full responsibility for it. Generative AI was not used to produce or alter any primary data, and is not listed as an author. The machine-learning methods described in Methods (Geneformer, scGPT, AlphaFold-Multimer, gradient boosting with SHAP) are research tools applied to the data and are distinct from the assistance declared here.

Data and code availability

This study analysed only publicly available, previously published human datasets under the accessions listed in Methods; no new human or animal experiments were performed and no new ethical approval was required. Versioned analysis code, a pinned software environment, random seeds, automated numerical checks, compact derived tables and the submitted manuscript snapshot are publicly available at `https://github.com/ArleneHarrison/arid1a-baf-collateral-dependency` (release v1.0.5) and archived at Zenodo under the version-independent concept DOI (`https://doi.org/10.5281/zenodo.22218064`). Raw source datasets are not redistributed and remain available from the persistent accessions and URLs listed in the repository data manifest. No claim in the manuscript depends on data available only on request.

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